

Debs Maia Silva, Ph.D.

Senior Manager, Data Science (GenAI Reference)

Capital One — Compliance & Ethics

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WORK EXPERIENCE

- **Capital One** McLean, VA
Manager, Data Science (GenAI Tech Lead) 07-2025 – Present
 - **Team Leadership & Management:** Lead and mentor a team of **5 Data Scientists**, serving as the GenAI SME for Compliance & Ethics. Manage the full talent lifecycle, including **technical hiring, performance management, and career coaching**.
 - **Strategic GenAI Delivery (Tester Assist):** Spearheaded the "Tester Assist" initiative from scoping to production. Secured **senior leadership buy-in and budget allocation** for automated natural-language data requests and GenAI-generated executive reporting, reducing reporting cycles from weeks to hours.
 - **Multimodal ML Orchestration:** Directed the cross-functional delivery (2 tech teams, 1 product team) of a production **Multimodal LLM pipeline**. Scaled compliance testing from a 60-document manual cap to **millions of records**, overseeing the project from MVP through feature expansion.
 - **Technical Lead (GenAI Code Synthesis):** Served as Tech Lead for a first-of-its-kind **English-to-Code generator** that translates legal/regulatory attributes into Python/SQL. Eliminated the primary manual bottleneck in risk testing, significantly accelerating time-to-market for compliance products.
- **Capital One** McLean, VA
Principal Associate - Data Science 06-2021 – 07-2025
 - **NLP & Risk Management:** Deployed Capital One's first production **LLM application** in a regulated environment. Built RAG-based search and automated evaluation systems for enforcement action summarization, establishing core guardrails and internal ML controls.
 - **Operations Optimization:** Engineered Deep Learning/NLP models to detect regulatory failures in call center audio/text. Achieved a **40% accuracy improvement** and cut analyst review time by **50%**.
 - **Fraud ML Lifecycle:** Developed an in-house debit fraud detection model using **PyTorch and Spark** on AWS. Managed the end-to-end lifecycle from feature engineering to stakeholder delivery, replacing a high-cost vendor system.
- **Purdue University** West Lafayette, IN
Graduate Research Assistant — Ph.D. Candidate 08-2017 – 05-2021
 - **High-Stakes Predictive Modeling:** Developed complex Bayesian additive regression trees and ensemble models for large-scale environmental datasets. Published as first author in **Nature Communications**; research featured in *Phys.org* and *Climate Central*.
- **Capital One** McLean, VA
Ph.D. Data Science Intern 06-2020 – 08-2020
 - **Graph Analytics:** Applied network embeddings and graph algorithms to identify fraudulent transaction clusters.

EDUCATION

- **Purdue University** West Lafayette, IN
*Ph.D. Purdue University College of Engineering (Focus: ML and Data Science), **Ross Fellow***
- **Campinas University (UNICAMP)** Campinas, Brazil
B.S. Civil Engineering; Minor in Computer Engineering

TECHNICAL AND LEADERSHIP SKILLS

- **Leadership:** Team Management, Strategic Roadmap Planning, Stakeholder Management, Budgeting, Technical Hiring, Cross-Functional Leadership (Tech/Product)
- **GenAI & ML:** LLMs, RAG, Transformer Architectures, Computer Vision, NLP, PyTorch, TensorFlow, Scikit-learn, MLOps, AI Evaluation Frameworks
- **Cloud & Data:** AWS (S3, SageMaker), Spark, SQL, Python, R, Vector Databases, Snowflake

PUBLICATIONS (SELECTED)

- **D. Maia-Silva** et al. "The role of humidity in electricity demand," *Nature Communications*, 2020.
- **D. Maia-Silva** et al. "The Goldilocks Zone in Cooling Demand," *AGU Earth's Future*, 2022.

PRESENTATIONS & LEADERSHIP PRESENCE

- **SECON (2025)**: Keynote - Transforming Regulatory Compliance with Generative AI.
- **MAC (2023–2025)**: Lead Panelist on AI Evaluation and Synthetic Data Generation.
- **Rewriting The Code / SHPE**: Recurring speaker on AI Engineering and Career Leadership for Women in Tech.

HONORS AND AWARDS

- Capital One **Patent**: Automated Compliance Hub Platform *2025*
- **Outstanding Research Award** – Purdue College of Engineering *2020*
- **Ross Fellowship** – Purdue College of Engineering *2017–2021*

VOLUNTEERING & MENTORSHIP

- Internship Site Lead – Capital One Data Science (Mentored 15+ interns over 3 years) *2022–2025*
- Founder, "Vem ser Dev": Data Science education platform (30k+ followers) *2020–Present*